# LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

**ANSWER KEY & MARKING SCHEME**

## Ch 6 Test: Control and Coordination

*Chapter / Topic: Control and Coordination*

*For teacher use only. Do not distribute to students.*

### SECTION A

**Q1.** Which of the following is a plant hormone? **[1]**

**Answer:** (b) Cytokinin

**Q2.** The gap between two neurons is called a: **[1]**

**Answer:** (c) synapse

**Q3.** The brain is responsible for: **[1]**

**Answer:** (d) all of the above

**Q4.** Which hormone regulates the blood sugar level? **[1]**

**Answer:** (a) Insulin (secreted by the pancreas)

**Q5.** The bending of a plant shoot towards light is called: **[1]**

**Answer:** (d) phototropism

**Q6.** **Assertion (A):** Iodised salt is advised in our diet. **Reason (R):** Iodine is needed by the thyroid gland to make thyroxin. **[1]**

**Answer:** (a). A deficiency of iodine can cause goitre.

### SECTION B

**Q7.** What is a reflex action? Give two examples. **[2]**

**Answer:** A sudden, automatic response to a stimulus that happens without conscious thinking. Examples (any two): pulling the hand away from a hot object, blinking when a bright light falls on the eyes, sneezing, knee-jerk.

- Definition: 1
- Two examples: ½ + ½

**OR, answer:** The leaves fold up and droop when touched. Cells change shape by changing the amount of water in them, and there is no growth involved. The bending of a shoot towards light is a growth-related movement: cells on the shaded side grow longer because of auxin.

- Touch-me-not: water/cell shape change, no growth: 1
- Shoot: growth-related movement: 1

**Q8.** Why is adrenaline called the 'emergency hormone'? Give two of its effects on the body. **[2]**

**Answer:** It is secreted by the adrenal glands directly into the blood when we are scared or in danger, and it prepares the body to respond. Effects (any two): the heart beats faster, sending more oxygen to the muscles; blood supply to the digestive system and skin is reduced; the breathing rate increases.

- Secreted in emergency situations: 1
- Two effects: ½ + ½

### SECTION C

**Q9.** Answer the following about a neuron: **[3]**

(i) At the dendritic tip (dendrites). [1]

(ii) From the dendrite to the cell body, and then along the axon to its end. [1]

(iii) At the end of the axon, the electrical impulse sets off the release of chemicals. These cross the gap (synapse) and start a similar electrical impulse in the dendrite of the next neuron. [1]

**Q10.** Name the plant hormone that: **[3]**

(i) Cytokinin [1]
(ii) Abscisic acid [1]
(iii) Auxin [1]

## SECTION D

**Q11.** Answer the following: **[5]**

(i) (a) Pituitary gland: deficiency of growth hormone in childhood. (b) Pancreas: insulin not secreted in proper amounts. (c) Thyroid gland: thyroxin, due to iodine deficiency. [3]

- Each gland + hormone: 1

(ii) The timing and amount of hormone released must be precise, and feedback mechanisms regulate this. Example: when blood sugar rises, cells of the pancreas detect it and secrete more insulin. As the sugar level falls, insulin secretion is reduced. [2]

- Regulates timing/amount: 1
- Insulin example: 1

**OR, answer:** (a) Forebrain: the main thinking part, with areas for sensory information, voluntary actions, hunger and so on. Midbrain: controls many involuntary actions. Hindbrain: the cerebellum controls posture and balance, and the medulla controls blood pressure, salivation and vomiting. (b) The brain sits inside a bony box (the skull/cranium) and is surrounded by a fluid-filled balloon that acts as a shock absorber.

- (a) each region with function: 1
- (b) skull 1, fluid/shock absorption 1

## SECTION E

**Q12.** A student placed a potted plant near a window for a week. The shoot bent towards the window, while the roots grew downwards. The student read that a… **[4]**

(i) Phototropism (positive). [1]
(ii) Auxin, made at the shoot tip. [1]
(iii) Auxin diffuses to the shaded side, where the cells grow longer than on the lit side, so the shoot bends towards light. Roots show positive geotropism (they grow downwards). [2]

- Shaded side cells elongate more: 1
- Positive geotropism: 1